M2 Quantitative Finance

Stochastic Modelling/Control in Finance

Dealing with the Inventory Risk: A Solution to the Market Making Problem

Stochastic Control Approach to Market Making

Project based on the paper by Guéant, Lehalle and Fernandez-Tapia

Aya El Moudden   Aymen Imad   Johanna Rakotovao

Academic year 2025–2026

Contents

Part I – Motivation and Model Setup

1 Introduction: article, motivation and course connection

The project consists in studying a research paper in stochastic control. According to the course guidelines, the analysis should cover three complementary dimensions: the modelling motivation and the origin of the paper in the literature, selected theoretical arguments or proof sketches, and a numerical implementation that can be compared with the paper's results, even if the reproduced figures do not coincide exactly.

The selected paper is *Dealing with the Inventory Risk: a Solution to the Market Making Problem* by Gueant, Lehalle and Fernandez-Tapia [1]. It is directly connected to the course chapter on *Optimal Dealing Under Constraints*: in that chapter, market makers are introduced as agents who provide liquidity, quote bid and ask prices, and have to manage both cash and inventory constraints [2]. The same chapter distinguishes liquidity takers, liquidity providers and market makers. This distinction is important here because the paper focuses on a liquidity-providing agent whose main risk is the accumulation of a long or short inventory.

The topic also fits the broader financial-markets part of the course, where classical frictionless assumptions are relaxed. In particular, the liquidity-risk course emphasizes that realistic trading models may involve transaction costs, price impact, discrete trading decisions and state constraints [3]. The model studied here is simpler than a full market microstructure model, but it already introduces a key friction: executions depend on where the market maker places her quotes.

How this part fits into the full project

Part 1	**Part 2**	**Part 3**
Motivation, course link, model variables and assumptions	HJB formulation, Bellman principle, change of variables and key proof ideas	Numerical solution, figures, sensitivity analysis and comparison with the paper

2 Market making and inventory risk

A market maker continuously quotes two prices:

- a **bid price**, at which she is willing to buy the asset;
- an **ask price**, at which she is willing to sell the asset.

Her natural source of revenue is the bid-ask spread. If she buys at the bid and sells at the ask, she earns a margin on the round trip. This explains why liquidity provision can be profitable.

The difficulty is that executions are random. The market maker may buy several times before selling back, or sell before being able to buy back. She then holds a positive or negative inventory. This inventory is exposed to fluctuations of the reference price: a long inventory loses value when the price decreases, while a short inventory loses value when the price increases. This is the *inventory risk* problem.

> **Main economic trade-off**
>
> The market maker can quote close to the reference price to increase the probability of execution, or quote farther away to improve revenue per trade and compensate for inventory risk. The optimal strategy must balance execution frequency, spread capture and inventory control.

This is a stochastic control problem because the market maker does not choose one static quote. She dynamically controls the bid and ask quote distances as time, executions and inventory evolve.

3 Positioning of the paper

The paper builds on the Avellaneda-Stoikov framework, itself rooted in the earlier market making model of Ho and Stoll. The authors study a high-frequency market maker operating on a single asset. The reference price follows a Brownian dynamics, executions are modelled by point processes, and execution intensities depend on the distance between the quotes and the reference price.

The paper has two important modelling choices:

- **inventory limits** are imposed, so the market maker cannot take arbitrarily large long or short positions;
- **CARA utility** is used, so terminal P&L risk is explicitly penalized.

The main mathematical contribution of the paper, developed in the theoretical part of this report, is the transformation of the nonlinear HJB system into a finite-dimensional linear system of ordinary differential equations. This reduction relies on the combination of CARA utility and exponential execution intensities. It makes the computation of optimal quotes more tractable than a direct numerical resolution of the full nonlinear HJB system.

4 Model setup

4.1 Probability space, information and horizon

We work on a filtered probability space

$$(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$$

satisfying the usual conditions. The filtration $(\mathcal{F}_t)_{t\geq 0}$ represents the information available to the market maker through time. The market maker acts over a finite horizon $[0,T]$, and her quoting strategy must be adapted to the information available at each time.

The core state variables are

$$(t, X_t, q_t, S_t),$$

where t is current time, X_t is the cash process, q_t is the signed inventory, and S_t is the reference price. In the HJB formulation, the same state is written with generic variables (t, x, q, s).

4.2 Reference price dynamics

The reference price, interpreted as a stylized mid-price or fair price, is denoted by S_t. In the baseline model, it follows an arithmetic Brownian motion:

$$dS_t = \sigma\, dW_t, \tag{1}$$

where W_t is a standard Brownian motion and $\sigma > 0$ is the instantaneous volatility.

This price process is intentionally simple. Over short horizons, the purpose is not to model the entire price formation mechanism, but to isolate the effect of price uncertainty on the value of the market maker's inventory. The paper later discusses extensions with drift and market impact, but the baseline setting starts from this exogenous Brownian reference price.

4.3 Bid/ask quotes and control variables

The market maker posts a bid quote S_t^b and an ask quote S_t^a. The controls are not the prices directly, but the distances from the reference price:

$$\delta_t^b = S_t - S_t^b, \qquad \delta_t^a = S_t^a - S_t. \tag{2}$$

Equivalently,

$$S_t^b = S_t - \delta_t^b, \qquad S_t^a = S_t + \delta_t^a. \tag{3}$$

Small quote distances mean that the market maker quotes close to the reference price. This increases the chance of execution but reduces revenue per trade. Large distances improve the price obtained when a trade occurs, but make execution less likely.

The admissible controls belong to a set $\mathcal{A}$ of predictable quote-distance processes. Economically, predictability means that the market maker cannot use future information when choosing quotes. The controls are also restricted by the inventory bounds: when inventory is already at the upper or lower limit, the market maker cannot choose a quote that would increase the position further.

4.4 Execution processes and intensities

Executions are represented by two point processes:

- N_t^b counts executions at the bid, hence purchases by the market maker;
- N_t^a counts executions at the ask, hence sales by the market maker.

The paper assumes that these processes are independent of the Brownian motion driving the reference price. Each execution changes the inventory by one unit.

The execution intensities depend on the distance between the quote and the reference price. The exponential specification is

$$\lambda^b(\delta^b) = Ae^{-k\delta^b}, \qquad \lambda^a(\delta^a) = Ae^{-k\delta^a}, \tag{4}$$

where $A > 0$ and $k > 0$ statistically characterize the liquidity of the asset: A is the baseline execution intensity, while k controls the exponential decay of the execution intensity with respect to the quote distance.

The interpretation is simple: the closer the quote is to the reference price, the larger the execution intensity. Conversely, a quote farther away gives a better price to the market maker if it is executed, but it is executed less frequently.

4.5 Inventory process

The market maker's inventory is the signed number of shares held at time t:

$$q_t = N_t^b - N_t^a. \tag{5}$$

A bid execution increases q_t by one unit because the market maker buys the asset. An ask execution decreases q_t by one unit because the market maker sells the asset.

The inventory is bounded:

$$q_t \in \{-Q, -Q+1, \dots, Q-1, Q\}.$$

At $q = Q$, the market maker stops posting bid quotes because buying more would increase the already long position. At $q = -Q$, she stops posting ask quotes because selling more would increase the already short position. These bounds represent risk limits and make the control problem well posed under inventory constraints.

4.6 Cash process

The cash process X_t records the cash generated by executed trades. When an ask order is executed, the market maker sells at price $S_t + \delta_t^a$ and cash increases. When a bid order is executed, she buys at price $S_t - \delta_t^b$ and cash decreases. Therefore,

$$dX_t = (S_t + \delta_t^a)dN_t^a - (S_t - \delta_t^b)dN_t^b. \tag{6}$$

The terminal mark-to-market wealth is

$$X_T + q_T S_T, \tag{7}$$

that is, final cash plus the value of the remaining signed inventory evaluated at the reference price.

4.7 CARA utility and objective function

The paper uses a CARA exponential utility function:

$$U(w) = -\exp(-\gamma w), \qquad \gamma > 0, \tag{8}$$

where γ is the absolute risk-aversion parameter. The market maker solves

$$\sup_{(\delta_t^a),(\delta_t^b)\in\mathcal{A}} \mathbb{E}\left[-\exp\left(-\gamma(X_T + q_T S_T)\right)\right]. \tag{9}$$

This criterion shows that the market maker is not simply maximizing expected spread revenue. She is also penalizing risky terminal P&L generated by inventory exposure.

5 Notation and economic interpretation

Notation	Mathematical meaning	Economic interpretation
S_t	Reference price, $dS_t = \sigma dW_t$	Stylized mid-price or fair price of the asset
W_t	Brownian motion	Source of random price uncertainty
σ	Volatility coefficient	Intensity of price risk carried by inventory
S_t^b	Bid quote	Price at which the market maker buys
S_t^a	Ask quote	Price at which the market maker sells
δ_t^b	Bid distance $S_t - S_t^b$	Control on the bid side
δ_t^a	Ask distance $S_t^a - S_t$	Control on the ask side
N_t^b	Bid execution counting process	Number of purchases by the market maker
N_t^a	Ask execution counting process	Number of sales by the market maker
λ^b, λ^a	Execution intensities	Speed/probability of execution at the posted quotes
A	Baseline intensity level	Overall liquidity / execution frequency
k	Decay rate in $Ae^{-k\delta}$	Sensitivity of executions to quote distance
q_t	Signed inventory $N_t^b - N_t^a$	Long if positive, short if negative
Q	Inventory bound	Maximum allowed long or short position

X_t	Cash process	Cash generated by executed trades
$X_T + q_T S_T$	Terminal mark-to-market wealth	Final cash plus value of remaining inventory
γ	CARA risk-aversion coefficient	Strength of P&L risk penalization
T	Finite horizon	Final time at which terminal P&L is evaluated

6 Modelling assumptions and limitations

The baseline model is deliberately stylized. The reference price has no drift, order sizes are constant and normalized to one, execution intensities are exponential, and market impact is ignored at first. These assumptions are not meant to describe every detail of a real limit order book. Their role is to keep the stochastic control problem tractable and to make the inventory-risk mechanism explicit.

This modelling choice is useful for the rest of the project. Because of the CARA utility and the exponential form of execution intensities, the HJB system admits a change of variables that reduces it to a linear system of ordinary differential equations. The following parts build precisely on this structure: first by deriving the HJB equations, then by solving the reduced ODE system numerically and comparing the resulting optimal quotes with the paper's figures.

7 Conclusion

This first part introduces the market making problem studied in the paper. The market maker earns the bid-ask spread by providing liquidity, but random executions can create a long or short inventory that is exposed to price movements. The decision variables are the bid and ask quote distances, and the state variables are time, cash, inventory and reference price.

The model is directly linked to the course on optimal dealing under constraints. It provides the modelling layer needed for the next part: the Bellman equation and HJB formulation start from the state variables, controls, execution intensities, inventory dynamics, cash process and CARA objective introduced above.

Part II – Theoretical Analysis

8 Stochastic Control Formulation

8.1 The Market Making Problem

A market maker continuously provides liquidity to the market by posting two prices:

- a **bid price**, at which the market maker is willing to buy the asset;

- an **ask price**, at which the market maker is willing to sell the asset.

The objective of the market maker is to earn profits from the bid–ask spread while controlling the risk generated by inventory fluctuations.

8.2 State Variables

The dynamics of the system are described through three state variables.

The reference price follows

$$dS_t = \sigma dW_t.$$

The inventory process is denoted by q_t:

$$q_t > 0 \quad \Rightarrow \quad \text{long inventory},$$

$$q_t < 0 \quad \Rightarrow \quad \text{short inventory}.$$

The cash account generated by executed trades is denoted by X_t.

8.3 Control Variables

The market maker controls the bid and ask quotes through

$$\delta_t^b = S_t - S_t^b, \qquad \delta_t^a = S_t^a - S_t.$$

A small quote distance corresponds to a quote posted close to the mid-price and therefore to a high execution probability. Conversely, a larger quote distance decreases the execution probability but increases the profit obtained from a trade.

8.4 Optimization Problem

The market maker seeks to maximize the expected utility of terminal wealth. The value function is therefore defined as

$$u(t, x, q, s) = \sup_{\delta^a, \delta^b} \mathbb{E}\left[-e^{-\gamma(X_T + q_T S_T)}\right].$$

The quantity

$$X_T + q_T S_T$$

represents the mark-to-market value of the portfolio at terminal time.

The resulting optimization problem is driven by the fundamental trade-off

$$\text{Spread Profit} \qquad \Longleftrightarrow \qquad \text{Inventory Risk.}$$

Posting quotes further away from the mid-price increases the profit obtained per trade but reduces the execution probability. Posting more aggressive quotes increases execution probability but may lead to large and risky inventory positions.

9 Dynamic Programming Principle and HJB Equation

9.1 Dynamic Programming Principle

At every instant, the market maker chooses bid and ask quotes while accounting for:

- future price uncertainty;
- future order executions;
- future inventory risk.

The state variables of the control problem are

$$(t, x, q, s),$$

where t denotes time, x is the cash position, q is the inventory level, and s is the current reference price.

According to Bellman's principle, the optimal decision at time t combines the immediate gain generated by executed trades with the continuation value associated with future optimal decisions.

9.2 Hamilton–Jacobi–Bellman Equation

Applying dynamic programming leads to the Hamilton–Jacobi–Bellman equation.

For interior inventory states $|q| < Q$, the value function satisfies

$$\begin{aligned} 0 =& \partial_t u(t,x,q,s) + \frac{1}{2}\sigma^2 \partial^2_{ss} u(t,x,q,s) \\ &+ \sup_{\delta^b} \lambda^b(\delta^b) \Big[u(t, x - s + \delta^b, q+1, s) - u(t,x,q,s) \Big] \\ &+ \sup_{\delta^a} \lambda^a(\delta^a) \Big[u(t, x + s + \delta^a, q-1, s) - u(t,x,q,s) \Big]. \end{aligned}$$

The terminal condition is

$$u(T, x, q, s) = -\exp\big(-\gamma(x + qs)\big).$$

The first term,

$$\partial_t u,$$

represents the evolution of the value function over time.

The diffusion term

$$\frac{1}{2}\sigma^2 \partial_{ss}^2 u$$

comes directly from the Brownian dynamics of the reference price

$$dS_t = \sigma dW_t.$$

The bid contribution

$$\lambda^b(\delta^b)\Big[u(t, x - s + \delta^b, q + 1, s) - u(t, x, q, s)\Big]$$

represents the expected gain associated with a bid execution. When a buy order is executed, the inventory increases from q to $q + 1$ while the cash position decreases.

Similarly,

$$\lambda^a(\delta^a)\Big[u(t, x + s + \delta^a, q - 1, s) - u(t, x, q, s)\Big]$$

represents the contribution of a sell execution. In this case, the inventory decreases and the cash position increases.

The supremum operators reflect the fact that the market maker optimally chooses the bid and ask quotes at every instant.

Order arrivals are modeled through exponential intensities

$$\lambda(\delta) = Ae^{-k\delta}.$$

At the inventory boundaries $q = Q$ and $q = -Q$, one control disappears because the market maker is no longer allowed to increase the magnitude of the inventory position. At the upper inventory boundary $q = Q$, the market maker cannot post a bid quote, since a bid execution would increase the inventory beyond the admissible limit. The HJB equation becomes

$$\begin{aligned}0 =& \partial_t u(t,x,Q,s) + \frac{1}{2}\sigma^2 \partial_{ss}^2 u(t,x,Q,s)\\ &+ \sup_{\delta^a} \lambda^a(\delta^a)\Big[u(t,x+s+\delta^a,Q-1,s) - u(t,x,Q,s)\Big].\end{aligned}$$

At the lower inventory boundary $q = -Q$, the market maker cannot post an ask quote, since an ask execution would decrease the inventory below the admissible lower bound. The HJB equation becomes

$$\begin{aligned}0 =& \partial_t u(t,x,-Q,s) + \frac{1}{2}\sigma^2 \partial_{ss}^2 u(t,x,-Q,s)\\ &+ \sup_{\delta^b} \lambda^b(\delta^b)\Big[u(t,x-s+\delta^b,-Q+1,s) - u(t,x,-Q,s)\Big].\end{aligned}$$

9.3 Why is the HJB Equation Difficult to Solve?

The HJB equation is highly nonlinear.

First, the bid and ask controls appear inside supremum operators.

Second, the inventory process creates a coupling between neighboring inventory states through the transitions

$$q \longrightarrow q+1, \qquad q \longrightarrow q-1.$$

As a consequence, the value function at inventory level q depends on the value function at neighboring inventory levels.

Standard PDE methods therefore do not provide explicit solutions.

The main contribution of Guéant, Lehalle and Fernandez-Tapia is to exploit the structure of the model in order to transform this nonlinear PDE into a linear system of ordinary differential equations.

10 Main Contribution: Reduction of the HJB to Linear ODEs

The central question addressed by the paper is the following:

How can one solve the nonlinear HJB equation?

The key observation is that the model combines:

- a CARA utility function;
- exponential execution intensities.

These two ingredients interact particularly well and make it possible to derive a remarkable change of variables that linearizes the problem.

11 CARA Ansatz and Change of Variables

11.1 Main Transformation

The authors introduce the ansatz

$$u(t, x, q, s) = -e^{-\gamma(x+qs)} v_q(t)^{-\gamma/k}.$$

The quantity

$$x + qs$$

corresponds to the mark-to-market wealth of the market maker.

The exponential factor naturally appears because the utility function is CARA:

$$U(x) = -e^{-\gamma x}.$$

The remaining function $v_q(t)$ captures the effects of inventory dynamics, execution probabilities and time dependence.

11.2 Role of Exponential Intensities

Execution intensities satisfy

$$\lambda(\delta) = Ae^{-k\delta}.$$

This exponential form is essential because it combines naturally with the exponential utility function. Without this specific structure, the reduction to a linear system would not be possible.

11.3 Sketch of the Derivation

The motivation behind the ansatz is to separate the mark-to-market wealth component from the inventory dynamics.

Starting from

$$u(t, x, q, s) = -e^{-\gamma(x+qs)} v_q(t)^{-\gamma/k},$$

we compute the derivatives appearing in the HJB equation.

Since only the function $v_q(t)$ depends explicitly on time, differentiation yields

$$\partial_t u = -\frac{\gamma}{k}\frac{\dot{v}_q(t)}{v_q(t)}u.$$

Next, differentiating with respect to the price variable gives

$$\partial_s u = -\gamma q\, u,$$

and therefore

$$\partial_{ss} u = \gamma^2 q^2 u.$$

This second derivative is particularly important because it produces the quadratic inventory term appearing later in the ODE system.

Substituting these expressions into the HJB equation allows the common exponential factor

$$-e^{-\gamma(x+qs)}$$

to be factorized.

After simplification, the variables x and s disappear entirely from the equation.

The optimization problem is then reduced to a collection of equations indexed only by the inventory level q.

The maximization with respect to the bid and ask controls can now be performed explicitly. The first-order optimality conditions yield the optimal quote distances

$$\delta_b^*(t,q) = \frac{1}{k}\ln\left(\frac{v_q(t)}{v_{q+1}(t)}\right) + \frac{1}{\gamma}\ln\left(1+\frac{\gamma}{k}\right), \qquad q \neq Q,$$

and

$$\delta_a^*(t,q) = \frac{1}{k}\ln\left(\frac{v_q(t)}{v_{q-1}(t)}\right) + \frac{1}{\gamma}\ln\left(1+\frac{\gamma}{k}\right), \qquad q \neq -Q.$$

Substituting these optimal controls back into the HJB equation yields a finite-dimensional linear system of ordinary differential equations indexed by the inventory state.

12 Reduced Finite-Dimensional ODE System

After applying the change of variables, the nonlinear HJB equation becomes, for interior inventory states $|q| < Q$,

$$\dot{v}_q = \alpha q^2 v_q - \eta \left(v_{q-1} + v_{q+1}\right).$$

At the upper boundary $q = Q$, the bid-side transition is no longer admissible, so the equation becomes

$$\dot{v}_Q = \alpha Q^2 v_Q - \eta v_{Q-1}.$$

At the lower boundary $q = -Q$, the ask-side transition is no longer admissible, so the equation becomes

$$\dot{v}_{-Q} = \alpha Q^2 v_{-Q} - \eta v_{-Q+1}.$$

The terminal condition is

$$v_q(T) = 1, \qquad q \in \{-Q, \dots, Q\}.$$

The coefficients are given by

$$\alpha = \frac{k}{2}\gamma\sigma^2, \qquad \eta = A\left(1 + \frac{\gamma}{k}\right)^{-(1+k/\gamma)}.$$

12.1 Interpretation of the Coefficients

The coefficient

$$\alpha = \frac{k}{2}\gamma\sigma^2$$

originates from the diffusion term of the reference price.

Its dependence on σ reflects the fact that inventory becomes more risky when price fluctuations increase.

Its dependence on γ shows that inventory penalties become stronger for more risk-averse market makers.

The coefficient

$$\eta = A\left(1 + \frac{\gamma}{k}\right)^{-(1+k/\gamma)}$$

comes from the optimization of the execution terms.

It measures the effective intensity with which the inventory can move from one state to a neighboring state through order executions.

12.2 Interpretation of the ODE System

The term

$$\alpha q^2 v_q$$

represents the inventory-risk penalty.

As inventory magnitude increases, the market maker becomes more exposed to price fluctuations. The quadratic dependence on q therefore penalizes large inventory positions.

The neighboring terms

$$v_{q-1}, \qquad v_{q+1},$$

represent inventory transitions generated by executed orders.

Indeed,

- a buy execution changes inventory from q to $q+1$;
- a sell execution changes inventory from q to $q-1$.

12.3 Matrix Interpretation

The system can be written in matrix form as

$$\dot{v} = Mv, \qquad v(T) = \mathbf{1}.$$

The solution is

$$v(t) = e^{-M(T-t)}\mathbf{1}.$$

The matrix M is tridiagonal.

Its diagonal entries correspond to inventory-risk penalties, while its off-diagonal coefficients describe transitions between neighboring inventory states.

This representation highlights the close connection between inventory dynamics and finite-state Markov chains.

13 Characterization of Optimal Bid and Ask Quotes

Once the ODE system has been solved, explicit optimal quotes can be obtained.

The optimal bid quote is

$$\delta^{b*}(t,q) = \frac{1}{k}\ln\left(\frac{v_q(t)}{v_{q+1}(t)}\right) + \frac{1}{\gamma}\ln\left(1+\frac{\gamma}{k}\right), \qquad q \neq Q.$$

The optimal ask quote is

$$\delta^{a*}(t,q) = \frac{1}{k}\ln\left(\frac{v_q(t)}{v_{q-1}(t)}\right) + \frac{1}{\gamma}\ln\left(1+\frac{\gamma}{k}\right), \qquad q \neq -Q.$$

13.1 Inventory Management Mechanism

To understand these expressions, consider

$$\frac{v_q}{v_{q+1}}.$$

This ratio measures the marginal value associated with increasing inventory.

If inventory is already too large, the value of state $q+1$ becomes relatively small compared with the value of state q.

As a consequence, the bid quote moves further away from the mid-price, reducing the probability of additional purchases.

The same reasoning applies symmetrically to the ask quote.

At the upper inventory boundary $q = Q$, no bid quote is posted because a bid execution would violate the inventory constraint.

Similarly, at the lower inventory boundary $q = -Q$, no ask quote is posted.

13.2 Financial Interpretation

When inventory becomes positive, the market maker seeks to reduce exposure.

The ask quote becomes more aggressive, facilitating sales, while the bid quote becomes less attractive.

Conversely, when inventory becomes negative, the market maker seeks to buy back shares.

The optimal quotes therefore generate a natural mean-reversion mechanism that continuously pushes inventory back toward zero.

14 Asymptotic Behaviour

When the horizon becomes large,

$$\delta^{b*}(t,q) \longrightarrow \delta^{b*}_{\infty}(q).$$

The limiting regime is related to an eigenvalue problem associated with the matrix M.

Far from terminal time, the influence of the terminal condition becomes negligible and the strategy becomes almost stationary.

15 Closed-Form Approximation

The asymptotic bid quote admits the approximation

$$\delta_\infty^{b*}(q) \simeq \frac{1}{\gamma}\ln\left(1+\frac{\gamma}{k}\right)+\frac{2q+1}{2}\sqrt{\frac{\sigma^2\gamma}{2kA(1+\gamma/k)^{1+k/\gamma}}}.$$

This formula makes explicit the dependence of the optimal quotes on volatility, execution intensity, risk aversion and inventory.

16 Economic Interpretation of Parameters

16.1 Effect of Volatility

$$\sigma\uparrow \quad\Rightarrow\quad \text{spread}\uparrow.$$

Higher volatility increases inventory risk and therefore requires wider spreads.

16.2 Effect of Baseline Execution Intensity

$$A\uparrow \quad\Rightarrow\quad \text{spread}\downarrow.$$

Higher execution intensity reduces execution risk and allows tighter quotes.

16.3 Effect of Risk Aversion

$$\gamma\uparrow \quad\Rightarrow\quad \text{more conservative quotes.}$$

More risk-averse market makers place greater emphasis on inventory control.

16.4 Effect of the Decay Parameter

The parameter k controls how rapidly execution probabilities decrease as quotes move away from the mid-price.

Larger values of k imply a stronger sensitivity of order arrivals to quote distance.

17 Conclusion

The market-making problem can be formulated as a stochastic control problem.

The associated HJB equation is nonlinear and difficult to solve because of the interaction between inventory dynamics and execution controls.

The key contribution of Guéant, Lehalle and Fernandez-Tapia is the introduction of a change of variables exploiting both CARA utility and exponential execution intensities.

This transformation reduces the original nonlinear HJB equation to a finite-dimensional linear system of ordinary differential equations.

The resulting framework provides explicit optimal quotes, asymptotic approximations and a clear economic interpretation of the role played by volatility, liquidity, inventory and risk aversion.

Part III – Numerical Implementation and Discussion

18 Numerical Results

18.1 Objective

The objective of this numerical study is to implement the Avellaneda–Stoikov market making model in order to simulate the behavior of a market maker exposed to inventory risk. The purpose of this implementation is to analyze dynamically how optimal bid and ask quotes evolve under stochastic market conditions while controlling inventory exposure and maximizing profitability.

The model considers a market maker who continuously posts two quotes on the market. The first quote corresponds to the bid quote, which represents the buying price proposed by the market maker, while the second quote corresponds to the ask quote, which represents the selling price. These quotes are updated dynamically throughout the trading horizon according to both market conditions and inventory levels.

The main objective of the strategy is twofold. On the one hand, the market maker aims to maximize trading profits generated through the bid–ask spread and order executions. On the other hand, the strategy must control inventory risk in order to avoid excessive exposure to adverse market movements. Consequently, the optimal quotes are continuously adjusted depending on the current inventory held by the market maker.

The market price is modeled using a stochastic Brownian motion process defined by

$$dS_t = \sigma dW_t$$

where σ represents the market volatility and W_t denotes a standard Brownian motion. This stochastic process introduces random fluctuations in the asset price throughout the trading horizon and allows the simulations to reproduce realistic market uncertainty.

The numerical implementation of the model is divided into several stages. The first step consists in simulating the stochastic market price process using a discrete Brownian motion approximation. The second step involves the numerical resolution of the system of ordinary differential equations associated with the Avellaneda–Stoikov framework through matrix exponential methods. Finally, the optimal bid and ask quotes are computed dynamically for all inventory states and time steps, allowing the simulation of the complete market making strategy under inventory risk.

18.2 Model Parameters

In order to implement numerically the Avellaneda–Stoikov market making framework, several parameters are introduced to characterize both market dynamics and the behavior of the market maker. Each parameter plays a specific role in the model and directly influences the optimal trading strategy generated by the simulations.

Parameter	**Role**	**Economic Interpretation**
σ	Volatility	Measures market price fluctuations and controls the uncertainty of the stochastic price process. Higher volatility increases inventory risk and leads to wider optimal spreads.
γ	Risk aversion	Represents the sensitivity of the market maker to inventory risk. Larger values of γ force the strategy to reduce inventory exposure more aggressively.
A	Base intensity	Defines the average intensity of order arrivals in the market. A larger value of A implies more frequent executions.
k	Liquidity parameter	Controls the decay of execution intensities with respect to the distance from the mid-price. It reflects market liquidity conditions and impacts quote aggressiveness.
q	Inventory	Represents the number of assets currently held by the market maker. The inventory level directly affects the optimal bid and ask quotes.
T	Time horizon	Corresponds to the end of the trading period and defines the total simulation horizon over which the strategy is optimized.

Table 2: Main parameters of the Avellaneda–Stoikov model.

The objective of the market maker is to maximize the expected utility of terminal wealth while controlling inventory risk throughout the trading horizon. The optimization problem is formulated through an exponential utility function defined by

$$\max \mathbb{E}\left[-e^{-\gamma(X_T+q_T S_T)}\right]$$

where X_T denotes the terminal cash process, q_T represents the final inventory position, and S_T corresponds to the terminal stock price.

This optimization criterion introduces a direct trade-off between profitability and risk management. The exponential utility penalizes excessive inventory exposure and forces the market maker to dynamically adapt bid and ask quotes according to both inventory conditions and market uncertainty. Consequently, the optimal strategy continuously balances execution profitability with inventory stabilization during the entire trading period.

18.3 Assumptions and Simplifications

In order to implement the Avellaneda–Stoikov framework numerically and obtain stable simulations, several modeling assumptions and simplifications are introduced. These assumptions allow the numerical resolution of the model while preserving the main economic mechanisms associated with market making under inventory risk.

The first major assumption concerns order arrivals. Buy and sell market orders are assumed to follow independent Poisson processes. Consequently, executions occur randomly over time according to stochastic intensities. This assumption provides a tractable probabilistic framework for modeling market activity and reproduces the random nature of financial order arrivals.

The execution intensities are modeled using exponential functions defined by

$$\lambda(\delta) = Ae^{-k\delta}$$

where A represents the baseline execution intensity and k controls the sensitivity of executions with respect to the distance from the mid-price. This exponential structure implies that quotes placed closer to the mid-price have higher execution probabilities, while more distant quotes are executed less frequently.

Another important assumption is that the market maker can continuously adjust bid and ask quotes throughout the trading horizon. At each time step, the strategy updates the quotes dynamically according to the current inventory position and market conditions. This continuous adaptation mechanism constitutes one of the core principles of the Avellaneda–Stoikov framework.

Several simplifications are also introduced in the numerical implementation. First, the model does not include any direct market impact generated by the market maker's own trades. Consequently, executed orders do not affect the underlying market price. This assumption considerably simplifies the simulations and avoids introducing additional nonlinear feedback effects into the model.

Transaction costs are neglected in the implementation. The strategy therefore focuses exclusively on spread capture and inventory management without accounting for brokerage fees or exchange commissions. In addition, market liquidity is assumed to remain constant throughout the simulation horizon, implying that the execution parameters do not vary over time.

The model parameters are considered stationary during the simulations. In particular, volatility, execution intensities, and liquidity conditions remain fixed during the trading period. This assumption ensures numerical stability and simplifies the calibration process.

The calibration procedure is performed using a limited market window. Consequently, the estimated parameters represent local market conditions rather than long-term market dynamics. Some simulations also use simplified trade sizes in order to reduce computational complexity and facilitate the analysis of inventory dynamics.

The objective of these assumptions and simplifications is not to reproduce all the complexities of real financial markets, but rather to construct a numerically stable and computationally tractable framework capable of preserving the main economic mechanisms of market making. Despite these simplifications, the model remains sufficiently realistic to capture the interaction between profitability, execution intensity, inventory risk, and quote dynamics.

18.3.1 Optimal Quotes and Spread Surfaces

The first numerical experiment consists in computing the optimal bid quotes, optimal ask quotes, and optimal spread generated by the Avellaneda–Stoikov framework. The objective of this simulation is to analyze dynamically how the market maker adjusts quotes according to both inventory risk and the remaining trading horizon. The numerical implementation is based on the discretization of time and inventory spaces together with the numerical resolution of the Hamilton–Jacobi system associated with the optimization problem.

The inventory space is discretized between $-Q$ and Q with $Q = 30$, leading to 61 possible inventory states. The trading horizon is discretized into 600 time steps in order to obtain sufficiently smooth quote dynamics and stable numerical results throughout the simulation. The parameters used in the implementation are fixed to

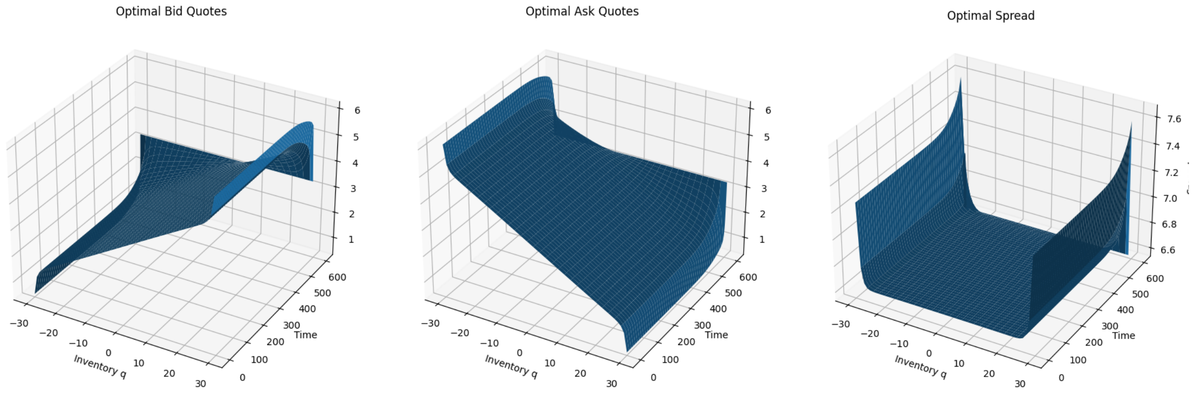
$$\sigma = 0.3, \qquad A = 0.9, \qquad k = 0.3, \qquad \gamma = 0.01, \qquad T = 600.$$

The numerical procedure begins with the construction of the tridiagonal matrix M

Once the matrix is constructed, the value functions associated with all inventory states are obtained numerically through the matrix exponential

$$v(t) = e^{-M(T-t)}\mathbf{1}.$$

The optimal bid and ask quotes are then computed dynamically using logarithmic ratios of neighboring value functions. This numerical approach allows the strategy to react continuously to inventory variations and stochastic market conditions.



(a) Optimal Bid Quotes (b) Optimal Ask Quotes (c) Optimal Spread

Figure 1: Optimal bid quotes, optimal ask quotes, and optimal spread surfaces generated by the Avellaneda–Stoikov model.

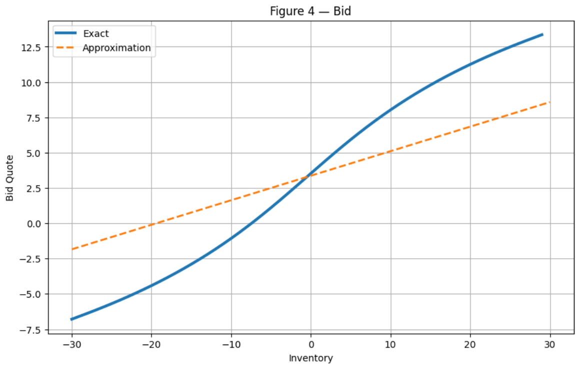
The numerical results clearly show that the optimal quotes depend simultaneously on both time and inventory. The bid quote surface increases progressively for positive inventory values while decreasing for negative inventories. Conversely, the ask quote surface decreases for positive inventories and increases for negative inventories. This asymmetric behavior reflects the inventory management mechanism implemented by the strategy.

When the inventory becomes strongly positive, the market maker aims to reduce exposure by encouraging selling activity. Consequently, bid quotes become less aggressive while ask quotes become more attractive to potential buyers. This mechanism accelerates inventory liquidation and prevents excessive long exposure. In contrast, when the inventory becomes strongly negative, the strategy encourages buying activity by increasing bid aggressiveness and reducing ask competitiveness in order to recover inventory neutrality.

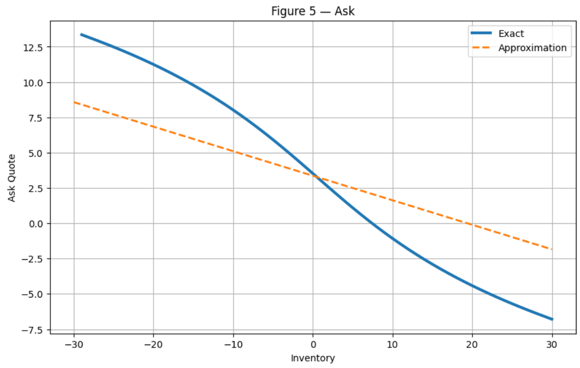
The optimal spread surface also exhibits important dynamic properties. Around zero inventory, the spread remains relatively stable and minimal because the inventory risk is limited. However, the spread widens significantly for extreme positive or negative inventory values. This widening effect reflects stronger inventory protection and a more conservative trading behavior under large exposure conditions.

The numerical surfaces remain smooth during the entire trading horizon, confirming the stability of the numerical implementation and the robustness of the discretization scheme. Moreover, the symmetry observed between positive and negative inventory regions is fully coherent with the theoretical structure of the Avellaneda–Stoikov framework.

These numerical results demonstrate that the optimal strategy dynamically balances profitability and inventory risk management. The simulations confirm that inventory-aware quote adjustments constitute the central mechanism through which the market maker stabilizes exposure while continuously providing liquidity to the market. The obtained behaviors are therefore fully consistent with the theoretical predictions of the Avellaneda–Stoikov model and validate the correctness of the numerical implementation.



(a) Exact vs Approximate Bid Quote



(b) Exact vs Approximate Ask Quote

Figure 2: Comparison between the exact numerical solution and the asymptotic approximation for bid and ask quotes.

The numerical results show that the asymptotic approximation follows correctly the global trend of the exact numerical solution for both bid and ask quotes. Around moderate inventory values, the approximation remains relatively close to the exact solution, which indicates that the asymptotic formulas capture correctly the dominant behavior of the optimal strategy.

However, the differences between the two approaches become significantly larger for extreme inventory values. The exact numerical solution exhibits stronger nonlinear behavior near the inventory boundaries, while the asymptotic approximation remains almost linear throughout the inventory range.

For large positive inventory positions, the exact bid quote increases more rapidly than the approximation. This reflects a stronger penalization of inventory accumulation in the exact solution and a more aggressive reduction of buying incentives. Similarly, for large negative inventory positions, the exact ask quote decreases more sharply than the approximation, encouraging stronger buying activity in order to restore inventory neutrality.

These observations demonstrate that the exact numerical solution captures more accurately the nonlinear inventory risk effects generated by large inventory exposure. Nevertheless, the asymptotic approximation remains sufficiently accurate for moderate inventories and provides a computationally efficient alternative for practical implementations.

Overall, this comparison highlights the trade-off between computational simplicity and numerical precision. The asymptotic formulas preserve the global structure of the optimal strategy while considerably reducing computational complexity, whereas the exact solution remains necessary for precise inventory risk management under extreme market conditions.

18.3.2 Price Dynamics and Inventory Process

The next numerical experiment studies jointly the evolution of the simulated stock price and the inventory process during the trading horizon. The objective of this simulation is to analyze how the market making strategy reacts dynamically to stochastic market fluctuations while maintaining inventory risk under control.

The stock price process is modeled using a Brownian motion discretization scheme. Starting from an initial market price S_0, the asset price evolves according to

$$S_{t+\Delta t} = S_t + \sigma\sqrt{\Delta t}\,\varepsilon_t,$$

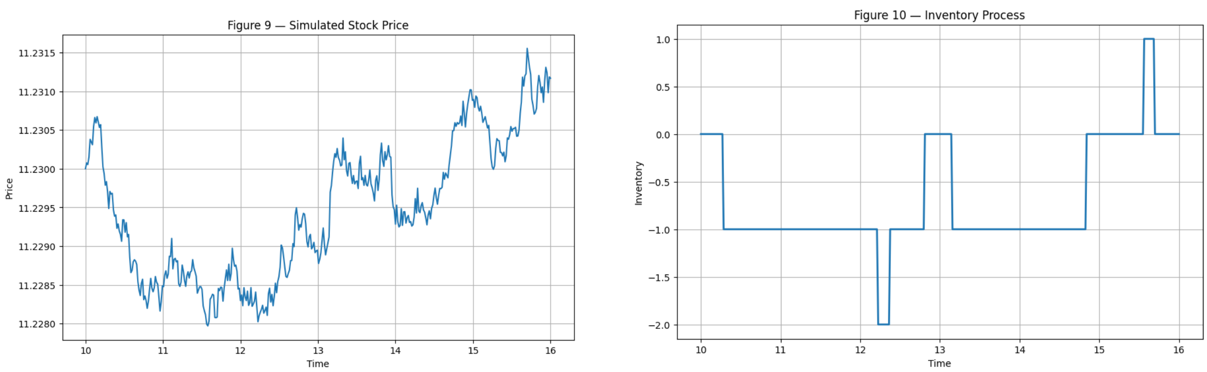
where

$$\varepsilon_t \sim \mathcal{N}(0,1).$$

This stochastic process generates random price fluctuations throughout the simulation horizon and provides a stylized representation of short-term market uncertainty.

At the same time, the inventory process evolves dynamically according to buy and sell order executions. The executions are generated using Poisson processes whose intensities depend on the optimal bid and ask quotes computed previously. Whenever a buy order is executed, the inventory increases by one unit, whereas a sell execution decreases the inventory by one unit.

The purpose of this joint simulation is to evaluate the stability of the strategy under stochastic market conditions and verify whether the inventory-aware mechanism successfully prevents excessive market exposure.



(a) Simulated Stock Price

(b) Inventory Process

Figure 3: Simulated stock price dynamics and inventory evolution during the trading horizon.

The simulated stock price exhibits continuous stochastic fluctuations that are fully consistent with Brownian motion dynamics. The trajectory evolves randomly around its initial value while preserving realistic short-term volatility behavior. Upward and downward movements appear continuously throughout the simulation horizon, reflecting the uncertainty naturally present in financial markets.

The inventory process evolves according to bid and ask order executions generated by the strategy. Despite the randomness of order arrivals and price fluctuations, the inventory remains globally close to zero during most of the simulation. This behavior demonstrates that the inventory-aware quoting mechanism successfully stabilizes the inventory position and prevents excessive accumulation of long or short exposure.

When the inventory becomes slightly positive or negative, the strategy dynamically adjusts bid and ask quotes in order to encourage opposite transactions and restore inventory neutrality. Consequently, inventory deviations remain temporary and relatively limited throughout the trading horizon.

The numerical results therefore confirm the effectiveness of the Avellaneda–Stoikov framework for controlling inventory risk under stochastic market conditions. The strategy continuously adapts to market fluctuations while maintaining stable inventory levels and preserving liquidity provision during the entire trading period.

18.3.3 P&L Comparison of Trading Strategies

This numerical experiment compares the profitability and stability of two different trading strategies implemented within the simulation framework. The first strategy corresponds to the optimal Avellaneda–Stoikov market making strategy, where bid and ask quotes are dynamically adjusted according to inventory risk. The second strategy corresponds to a naive strategy, where quotes are generated without inventory-aware optimization.

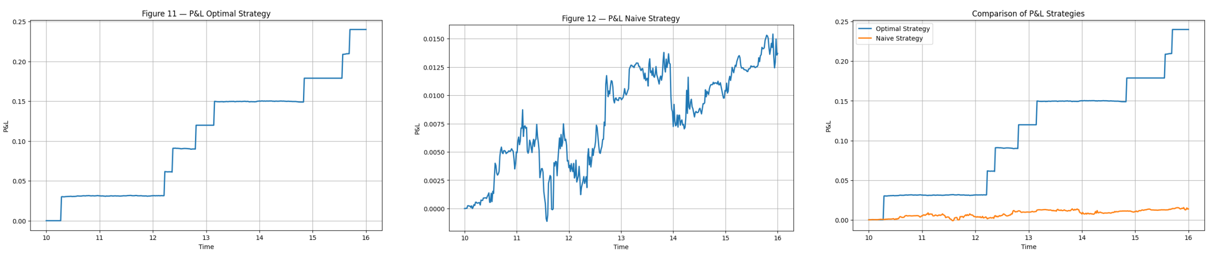
The objective of this comparison is to evaluate quantitatively the contribution of inventory management to profitability and risk control. For both strategies, the Profit and Loss process is computed continuously during the trading horizon by combining the cash account generated by executed trades with the marked-to-market value of the inventory position.

The P&L process is defined by

$$\text{P\&L}_t = X_t + q_t S_t,$$

where X_t denotes the cash process accumulated through executions, q_t represents the inventory process, and S_t corresponds to the market price at time t.

In the optimal strategy, the bid and ask quotes are continuously updated according to the current inventory state and market conditions. This dynamic adaptation allows the strategy to balance profitability and inventory risk throughout the trading period. In contrast, the naive strategy does not incorporate inventory-sensitive adjustments and therefore reacts less efficiently to stochastic market fluctuations.



(a) Optimal Strategy (b) Naive Strategy (c) Strategy Comparison

Figure 4: Comparison of P&L trajectories generated by the optimal and naive trading strategies.

The numerical results show that the optimal strategy generates a significantly higher and more stable P&L trajectory throughout the simulation horizon. The profit accumulation produced by the optimal strategy appears relatively smooth and monotonic, indicating efficient inventory management and stable execution dynamics.

In contrast, the naive strategy exhibits much stronger fluctuations and considerably lower profitability. The absence of inventory-aware quote adjustments leads to larger exposure to adverse market movements and less efficient execution behavior. Consequently, the naive strategy accumulates profits more slowly and remains more sensitive to stochastic market volatility.

The comparison between both trajectories clearly demonstrates the importance of dynamic inventory management in market making strategies. By continuously adapting bid and ask quotes according to inventory conditions, the Avellaneda–Stoikov framework reduces excessive exposure while improving spread capture efficiency.

The numerical simulations therefore confirm that inventory-aware quoting mechanisms improve simultaneously profitability and risk control. These results validate the efficiency of the Avellaneda–Stoikov model for optimal market making applications under stochastic market conditions.

18.3.4 Quotes and Executed Trades

This numerical experiment analyzes jointly the evolution of the market price, the optimal bid and ask quotes, and the executed trades generated during the simulation. The objective of this study is to verify the consistency between quote dynamics, execution mechanisms, and inventory management throughout the trading horizon.

The market price is simulated using the stochastic Brownian motion process introduced previously, while the optimal bid and ask quotes are computed dynamically using the Avellaneda–Stoikov framework. At each time step, execution probabilities are determined according to the bid and ask execution intensities

$$\lambda(\delta) = Ae^{-k\delta}.$$

Whenever a buy or sell market order arrives, a transaction is executed at the corresponding optimal quote. The simulation therefore reproduces dynamically the interaction between stochastic market prices, optimal quoting strategies, and executed trades.

The purpose of this experiment is to evaluate whether the generated quotes remain coherent with market conditions while preserving stable inventory control and realistic execution behavior.

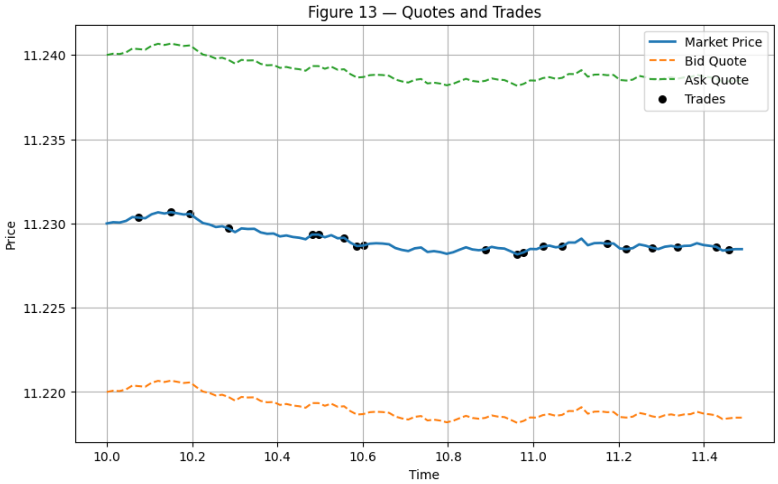


Figure 5: Market price, optimal bid and ask quotes, and executed trades during the simulation horizon.

The numerical results show that the bid and ask quotes evolve continuously around the market price throughout the trading horizon. The bid quote remains systematically below the market price, while the ask quote remains above it, preserving a strictly positive spread during the entire simulation. This behavior confirms the consistency of the quote generation mechanism implemented in the model.

The executed trades occur close to the optimal bid and ask quotes, indicating that the execution process remains coherent with the computed execution intensities. The distribution of transactions around the optimal quotes demonstrates that the strategy continuously adapts to market fluctuations while maintaining realistic execution dynamics.

The spread remains relatively stable during the simulation, although small variations appear due to inventory adjustments and stochastic market movements. When inventory deviations occur, the strategy dynamically modifies quote aggressiveness in order to encourage opposite transactions and restore inventory neutrality.

The numerical simulations therefore confirm that the Avellaneda–Stoikov framework generates realistic bid and ask quotes while preserving effective inventory risk management. The interaction between quote dynamics, stochastic price evolution, and executed trades remains stable throughout the trading horizon, validating the robustness of the numerical implementation.

The approximation errors between exact and approximate bid quotes are summarized below.

Metric	Value
MAE Bid	3.30
RMSE Bid	3.60
Maximum Error	4.95

Table 3: Approximation errors associated with bid quote estimation.

The relatively moderate values of the Mean Absolute Error and Root Mean Square Error confirm that the approximation remains globally accurate during the simulation. However, the maximum error indicates that larger deviations may still occur for extreme inventory positions where nonlinear inventory effects become stronger.

18.3.5 Sensitivity Analysis of the Spread

This numerical experiment investigates the sensitivity of the optimal spread with respect to the main parameters of the Avellaneda–Stoikov model. The objective of this analysis is to understand how market volatility, liquidity conditions, and risk aversion influence the behavior of the optimal quoting strategy and inventory management mechanism.

Three parameters are analyzed independently during the simulations:

$$\sigma \quad \text{(volatility)}, \qquad k \quad \text{(liquidity parameter)}, \qquad \gamma \quad \text{(risk aversion)}.$$

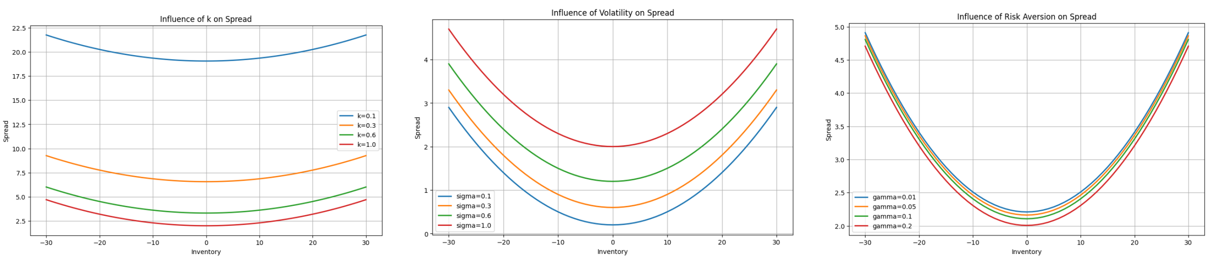
For each parameter configuration, the optimal spread is recomputed numerically for all inventory values while keeping the remaining parameters fixed. The spread is defined as the difference between the ask price and the bid price. Since

$$S^b(q,t) = S(q,t) - \delta^b(q,t), \qquad S^a(q,t) = S(q,t) + \delta^a(q,t),$$

we obtain

$$\text{Spread}(q,t) = S^a(q,t) - S^b(q,t) = \delta^a(q,t) + \delta^b(q,t).$$

The purpose of this study is to evaluate how the market maker dynamically adapts quote aggressiveness according to market uncertainty and inventory exposure.



(a) Influence of k (b) Influence of σ (c) Influence of γ

Figure 6: Sensitivity analysis of the optimal spread with respect to the liquidity parameter, volatility, and risk aversion.

The numerical results show that the spread remains minimal around zero inventory for all parameter configurations. This behavior is expected because the market maker faces lower inventory exposure near inventory neutrality and therefore requires less protection against adverse market movements.

The first sensitivity analysis concerns the liquidity parameter k. The simulations indicate that increasing k reduces the optimal spread. A larger value of k implies that execution intensities decay more rapidly with quote distance from the mid-price. Consequently, the market maker is encouraged to post more competitive quotes closer to the market price in order to maintain sufficient execution probabilities.

The second analysis focuses on market volatility σ. The results show that higher volatility significantly increases the spread for all inventory values. Since volatility amplifies market uncertainty and potential adverse price movements, the market maker widens the spread in order to protect against inventory risk and compensate for increased market exposure.

The third analysis investigates the influence of the risk aversion parameter γ. The numerical simulations reveal that larger values of γ strengthen inventory protection and generate wider spreads, especially for extreme inventory positions. A more risk-averse market maker reacts more aggressively to inventory deviations and therefore adjusts quotes more conservatively in order to reduce exposure.

The spread surfaces remain symmetric with respect to positive and negative inventory values, which confirms the consistency of the numerical implementation and the stability of the optimization framework.

The global influence of the different parameters on the spread is summarized in the following table.

Parameter	Effect on Spread
$\sigma \uparrow$	Spread increases
$k \uparrow$	Spread decreases
$\gamma \uparrow$	Stronger inventory protection

Table 4: Summary of parameter effects on the optimal spread.

Overall, the sensitivity analysis confirms that the Avellaneda–Stoikov framework dynamically adapts the optimal spread according to market conditions and risk preferences. The model successfully balances profitability and inventory risk management by adjusting quote aggressiveness in response to volatility, liquidity, and inventory exposure.

18.3.6 Calibration Using Real Market Data

The final numerical experiment consists in calibrating the Avellaneda–Stoikov model using real cryptocurrency market data collected from Coinbase. The objective of this calibration procedure is to estimate realistic market parameters and verify whether the theoretical framework is capable of reproducing actual market execution dynamics.

Due to geographical restrictions and API limitations encountered during the implementation phase, Coinbase market data were used instead of Binance data. Historical BTC–USD order book and transaction data were collected in order to estimate the execution intensity function and market volatility.

The calibration focuses primarily on the estimation of the execution intensity model defined by

$$\lambda(\delta) = Ae^{-k\delta},$$

where A denotes the baseline execution intensity and k controls the decay of execution probabilities with respect to the distance from the mid-price.

The calibration procedure begins by computing the empirical execution frequencies associated with different quote distances from the mid-price. An exponential regression is then performed numerically in order to estimate the parameters A and k. Simultaneously, the market volatility parameter σ is estimated directly from historical BTC–USD price variations.

The purpose of this experiment is to evaluate whether the Avellaneda–Stoikov framework remains coherent under realistic market conditions and whether the estimated parameters produce economically meaningful results.

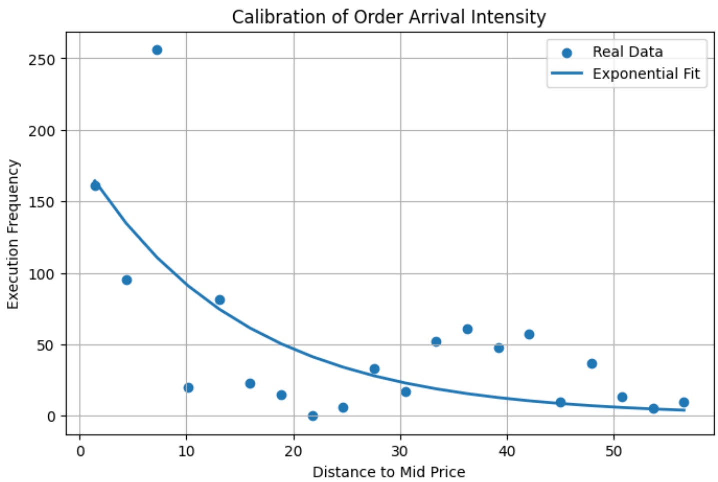


Figure 7: Calibration of execution intensities using real Coinbase BTC–USD market data.

The numerical results show that the exponential execution intensity model provides a satisfactory fit to the empirical Coinbase data. The execution frequency decreases progressively as the quote distance from the mid-price increases, which is fully consistent with the theoretical assumptions of the model.

The estimated parameters obtained through the calibration procedure are summarized below.

Parameter	Estimated Value
σ_{real}	1.20×10^{-5}
A	180.96
k	0.0678
Best Bid	78202.54
Best Ask	78202.55

Table 5: Calibration results obtained using Coinbase BTC–USD market data.

The estimated volatility remains relatively small over the calibration window, reflecting the short-term nature of the sampled market data. The value of the parameter A indicates a relatively high baseline execution intensity, while the estimated liquidity parameter k confirms the exponential decay of execution probabilities as quotes move away from the mid-price.

The calibrated bid and ask prices obtained from the Coinbase order book remain fully coherent with realistic BTC–USD market conditions. Moreover, the exponential fit reproduces correctly the global behavior of the empirical execution frequencies observed in the data.

These numerical results demonstrate that the Avellaneda–Stoikov framework is capable of capturing realistic market execution dynamics when calibrated using real financial data. Despite the simplifications introduced in the model, the calibration confirms that the framework preserves the essential mechanisms governing liquidity provision, execution probabilities, and inventory-aware market making strategies.

18.4 Comparison with the paper

The following table summarizes how the numerical implementation relates to the main qualitative results of the paper.

Result	**Paper**	**Our implementation**	**Comment**
Optimal bid quote surface	Bid quotes depend on inventory and time.	The bid quote surface reacts to inventory levels.	The bid becomes less aggressive when inventory is high.
Optimal ask quote surface	Ask quotes show the symmetric inventory effect.	The ask quote surface reacts in the opposite direction.	The ask becomes more aggressive when inventory is high.
Spread behaviour	The spread reflects inventory risk and execution risk.	The spread widens for extreme inventory values.	This is consistent with the inventory-risk mechanism.
Exact vs asymptotic approximation	The approximation is more accurate away from boundary effects.	Approximation errors are larger for extreme inventory values.	This is consistent with stronger nonlinear effects near inventory limits.
Coinbase calibration	Not part of the original numerical results.	Added as an illustrative extension.	It is not directly comparable to the paper's original figures.

Table 6: Comparison between the paper's qualitative results and our numerical implementation.

18.5 Conclusion

The numerical experiments carried out in this study show that the Avellaneda–Stoikov-type framework studied in the paper provides a relevant setting for analysing market making under inventory risk. Throughout the simulations, the strategy dynamically adapts bid and ask quotes according to market conditions and inventory exposure, helping the market maker maintain more balanced positions while continuously providing liquidity.

The results also highlight the importance of inventory-aware quote adjustments. Compared to simpler strategies, the optimal strategy generates more stable profits in the simulated setting and better controls inventory fluctuations. The sensitivity analysis further shows that volatility, liquidity, and risk aversion have a significant impact on spread dynamics and trading behaviour.

In addition, the calibration performed using Coinbase BTC–USD market data provides an illustrative example of how the exponential execution-intensity assumption can be fitted to real market data. Despite the simplifying assumptions introduced in the simulations, the calibrated parameters remain economically interpretable.

Overall, the numerical implementation reproduces the main qualitative mechanisms of the framework and provides useful simulations of market making activity under stochastic market conditions.

References

[1] O. Gueant, C.-A. Lehalle and J. Fernandez-Tapia, *Dealing with the Inventory Risk: a Solution to the Market Making Problem*, arXiv:1105.3115v5, 2012.

[2] E. Chevalier, *Stochastic Modelling/Control in Finance 3 - Optimal Dealing Under Constraints*, M2QF ENSIIE / Universite Paris-Saclay, course slides.

[3] V. Ly Vath, *Stochastic Modelling/Control in Finance 3 - Liquidity Risk and Optimal Portfolio*, M2QF ENSIIE / Universite Paris-Saclay, course slides.

[4] M2QF, *Stochastic Modelling/Control in Finance - Introduction, Corporate Finance Modelling, Financial Markets*, course slides.